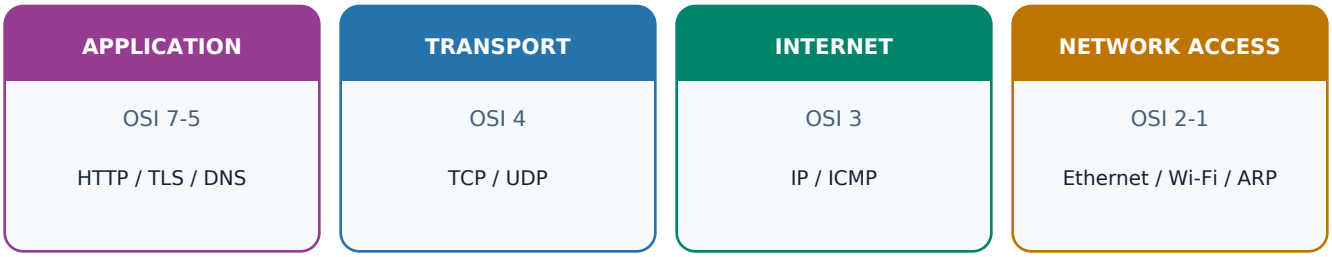
TCP/IP Packet Journey | Study Guide

A companion to MockTIA_Topic14_TCPIP_Packet_Journey.html. Use it before, during, and after the four interactive missions.

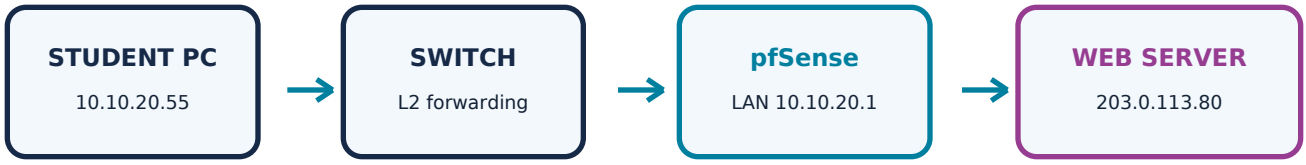
01 | Understand the four-layer model



Sending: application down to network access | Receiving: network access up to application

TCP/IP layer	Main job	OSI mapping and examples
Application	Services and data for applications	OSI 7-5 | DNS, HTTP(S), TLS, SSH
Transport	End-to-end connections, ports and delivery	OSI 4 | TCP segments; UDP datagrams
Internet	Logical addressing and routing	OSI 3 | IPv4, IPv6, ICMP
Network Access	Local-link framing and signaling	OSI 2-1 | MAC, Ethernet, Wi-Fi, ARP

The simulated lab network



PC MAC 02:10:20:00:00:55

TCP 443 | HTTPS (TLS)

IMPORTANT The domain training.example and the public IP addresses 203.0.113.80 and 198.51.100.10 are documentation examples. The browser lab simulates this packet flow; it does not send real network packets.

Remember: OSI is a conceptual teaching and troubleshooting reference. TCP/IP describes the protocols used in real networks. A four-layer TCP/IP diagram is a model, not four separate devices.

Mission 01 | Follow the Packet

A student at 10.10.20.55/24 opens https://training.example. Trace the five scored packet checkpoints in the same sequence as the lab.

1. **A URL starts the request**
   The PC needs an IP address for the hostname; no DNS answer is cached.

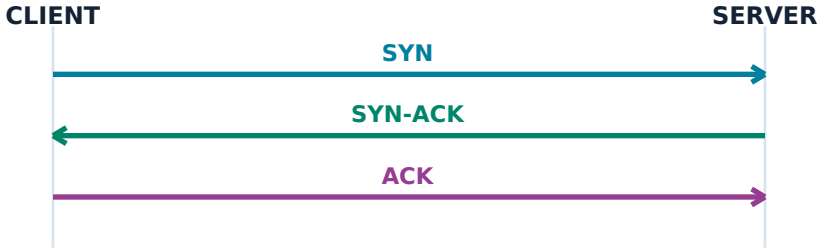
2. **ARP learns the next-hop MAC**
   To reach DNS/gateway 10.10.20.1, the PC learns 02:10:20:00:00:01.

3. **DNS returns an A record**
   UDP source 53014 to destination 53; training.example -> 203.0.113.80.

4. **TCP SYN leaves; pfSense performs PAT**
   10.10.20.55:51514 is translated to 198.51.100.10:62001.

5. **TCP, TLS and the return path**
   SYN-ACK -> ACK, TLS setup, encrypted HTTP; reply uses reverse NAT.

TCP three-way handshake (before TLS-protected HTTP)



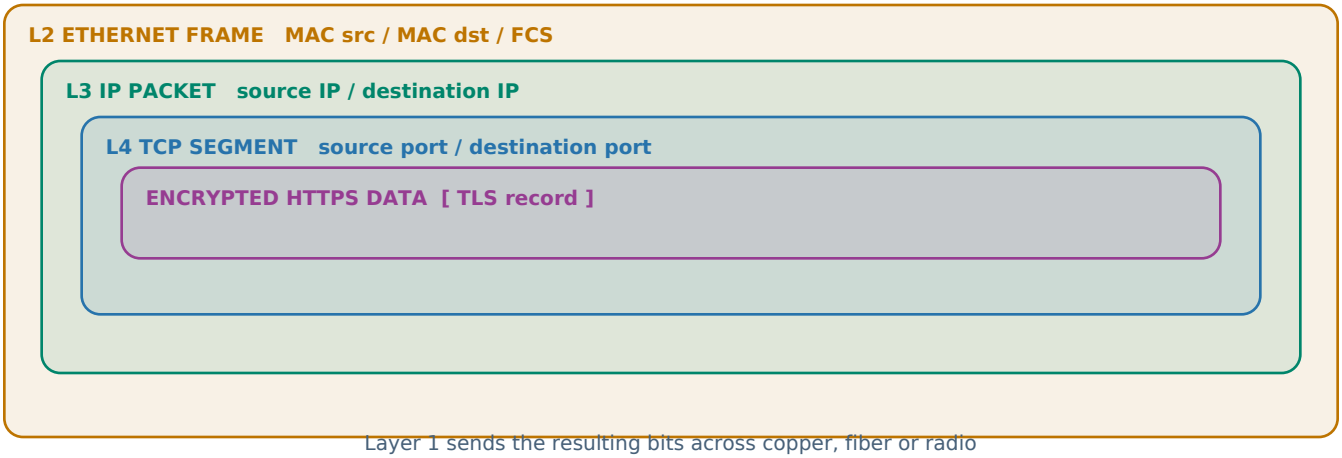
KEY DETAIL SYN → SYN-ACK → ACK establishes the TCP connection. Next, a TLS handshake establishes protection for application data. The server response returns through the firewall state/NAT mapping.

Think about these questions

Why does the PC use the gateway MAC instead of the remote web server MAC? Which values may change at a router: Layer 2 addresses, IP addresses, or both? Under what condition does NAT change the IP address?

Encapsulation | Envelopes and Encryption

Think of an application message as a letter. Each network layer adds delivery information around it. Encryption locks the message; encapsulation packages it for delivery.



Layer 1 sends the resulting bits across copper, fiber or radio

What is added	Example in this mission	Envelope analogy
TLS protection	HTTPS application data is encrypted	Lock the contents of the letter
TCP header	Source port 51514; destination port 443	Name the correct office/service
IP header	10.10.20.55 -> 203.0.113.80	Write logical destination address
Ethernet frame	PC MAC -> pfSense next-hop MAC	Label the local delivery envelope
Physical signals	Bits on the wire or radio	Carry the envelope on the road

ON EACH HOP Routers remove the incoming Layer 2 framing and create new framing for the next link. Source/destination MACs usually change. IP source/destination typically remain end-to-end, except where NAT or other network functions change them.

Encryption is not the same thing as encapsulation

TLS: protects application data for HTTPS. WPA2/WPA3: protects Wi-Fi links. IPsec: can protect IP-layer traffic. Even with HTTPS, intermediate equipment generally still needs routing metadata, such as visible IP addresses and transport information, to forward traffic.

Receiving direction: the server or client decapsulates information from the lower layers upward. A receiving HTTPS endpoint decrypts the TLS-protected contents after successful cryptographic processing.

Mission 02 | Windows Command Center

The lab terminal is simulated, but the five commands are real Windows diagnostic tools. Know what each result proves - and what it does not prove.

Enter exactly in the simulation	What to look for	Why it matters
`ipconfig /all`	IPv4 10.10.20.55; mask /24; gateway/DNS 10.10.20.1	Checks host configuration and assigned DNS
`arp -a`	Gateway IP mapped to MAC 02-10-20-00-00-01	Shows a local next-hop neighbor entry
`route print`	0.0.0.0/0 route via 10.10.20.1	Verifies off-subnet forwarding decision
`nslookup training.example`	A record maps name to 203.0.113.80	Checks DNS name-to-address resolution
`Test-NetConnection training.example -Port 443`	TcpTestSucceeded : True	Tests TCP reachability to HTTPS port 443

Simple diagnostic sequence

1. **Is my PC configured correctly?**
   Review IP, mask, gateway, and DNS using ipconfig /all.
2. **Can I reach my next hop?**
   Inspect ARP and routing; test gateway reachability.
3. **Does the hostname resolve?**
   Use nslookup and confirm the resolver response.
4. **Can my transport connection succeed?**
   Test TCP 443 rather than relying on ping alone.

TECHNICIAN TIP A successful ping does not prove that a website is reachable on TCP 443. DNS, a firewall policy, TLS, or the destination service may still prevent access.

Mission 03 | Troubleshooting Tickets

Each ticket is a first-pass diagnosis. The evidence directs you to the most likely failed component, but a real technician verifies the hypothesis.

Ticket and clue	Likely area	Where to start
A — IP works; hostname fails	DNS resolver / DNS configuration	Check nslookup, resolver status and configured DNS server
B — LAN works; off-subnet fails	Default gateway / routing	Check route print and the 0.0.0.0/0 route
C — Ping works; HTTPS fails	TCP 443 filtering / firewall rule	Test TCP 443 and inspect firewall policy
D — Outgoing SYN retains private WAN source	Outbound NAT / PAT	Check pfSense outbound NAT and state table
E — ARP for gateway unanswered	VLAN / Layer 2 path	Inspect switchport VLAN, tagging, link and ARP

pfSense and packet-capture tools in the lab

pfSense menu	Question it helps answer
Diagnostics → DNS Lookup / Ping	Do names resolve? Is a host reachable by ICMP?
Diagnostics → Routes / States	Which route is selected? Is a connection being tracked?
Firewall → Rules / NAT → Outbound	Is traffic permitted? Is source NAT correct?
Diagnostics → Packet Capture	Where do SYN, DNS, ARP, or ICMP packets appear?

Useful Wireshark display filters

`dns`	`arp`	`icmp`
`tcp.port == 443`	`tcp.flags.syn == 1`	`ip.addr == 10.10.20.55`

WATCH FOR A TCP SYN without a SYN-ACK can indicate a filtering, routing, service, or return-path issue. Capture at more than one point to isolate which device drops or fails to forward the packet.

Mission 04 | Certification Prep and Vocabulary

Use these points to prepare for the five-question assessment and to connect the Packet Journey exercise to CompTIA Network+ and Security+.

Question topic	Must-know answer / explanation
TCP/IP Internet layer mapping	Corresponds most closely to OSI Layer 3 (Network).
TCP connection setup	SYN → SYN-ACK → ACK is the TCP three-way handshake.
UDP behavior	Connectionless; it does not provide built-in reliable delivery.
Router drops packet with TTL = 0	Usually sends ICMP Time Exceeded to the sender.
HTTPS over TCP sequencing	TCP establishes its connection, then TLS establishes protection before encrypted HTTP request data is sent.

Vocabulary quick reference

Term	Plain-language meaning
DNS / A record	Turns a domain name into an IPv4 address.
ARP	Maps an IPv4 next-hop address to a MAC address on a local link.
MAC / IP	MAC: local-link address. IP: logical address used for routing.
Gateway	Router used to reach destinations outside the local subnet.
TCP / UDP	TCP: ordered, reliable byte stream. UDP: connectionless datagrams.
Port 443 / TLS	443: common HTTPS service port. TLS: encrypts/authenticates the connection.
NAT / PAT	Changes address/port information; state maps replies to the correct host.
Frame / Packet / Segment	Layer 2 / Layer 3 / Layer 4 units of data.
ICMP / TTL	Control and diagnostic messages / hop limit field.
VLAN	Logical separation of Layer 2 broadcast domains.

Completion checklist

☐ Five packet checkpoints ☐ Five Windows commands ☐ Five trouble tickets ☐ Five knowledge questions

The lab awards 5 points each for 20 challenges = 100 points. Enter your name in the HTML lab, complete the activities, and use Print / Save PDF in the lab itself to submit your scored completion report in Canvas. This PDF is a study aid, not the scored report.